



Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

Fundamentals of AI Ethics and Responsible AI

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Conducted by

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
6.0	1 July 2026	Legacy courseware release.	Dr Alfred Ang
7.0	9 September 2026	Aligned 2-day plan to the rebuilt 8-module deck and activity portfolio.	Dr Alfred Ang

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Course Overview

Fundamentals of AI Ethics and Responsible AI is a two-day, 16-hour WSQ course that develops the ability to identify responsible AI principles and apply an evidence-led governance framework to AI projects.

Learning Outcomes

- LO1: Identify AI ethics and governance principles.
- LO2: Apply an AI ethics and governance framework to AI projects.

Course Proposal Constructive Alignment

Learning Unit	Learning Outcome	TSC statements	Modules	Assessment
LU1 · Introduction	LO1 · Identify principles	K1, K2, A1	Day 1 09:00–17:00 excl. lunch · 7h	WA Q1–Q2; CS Q1
LU2 · Applying frameworks	LO2 · Apply framework	K3, K4, A2, A3	Day 1 17:00–18:00 + Day 2 09:00–16:00 excl. lunch · 7h	WA Q3–Q4; CS Q2–Q3

Daily Schedule

Day 1

Time	Duration	Topic / Activity	Method	Slides
09:00–09:30	30 min	Administration, introductions and AM attendance	Facilitated learning / activity	—
09:30–10:30	60 min	Module 1 · AI Ethics Foundations	Facilitated learning / activity	Slides 18–28
10:30–10:40	10 min	Tea break	Break	—
10:40–11:25	45 min	Activity 1 · Use-case and stakeholder map	Facilitated learning / activity	Slides 18–28
11:25–12:25	60 min	Module 2 · Responsible AI Principles	Facilitated learning / activity	Slides 29–39
12:25–13:10	45 min	Activity 2 · AI Code of Conduct	Facilitated learning / activity	Slides 29–39
13:10–14:10	60 min	Lunch break	Break	—
14:10–14:50	40 min	Module 3 · AI Governance Frameworks	Facilitated learning / activity	Slides 41–51
14:50–15:50	60 min	Activity 3 · Risk tiering and RACI	Facilitated learning / activity	Slides 41–51
15:50–16:00	10 min	Tea break	Break	—
16:00–17:00	60 min	Module 4 · AI Governance Lifecycle	Facilitated learning / activity	Slides 52–62
17:00–17:50	50 min	Activity 4 · Impact assessment and risk register	Facilitated learning / activity	Slides 52–62
17:50–18:00	10 min	Day 1 recap and PM attendance	Facilitated learning / activity	—

Day 2

Time	Duration	Topic / Activity	Method	Slides
09:00–09:10	10 min	Day 1 recap and AM attendance	Facilitated learning / activity	—
09:10–09:50	40 min	Module 5 · Fairness and Explainability	Facilitated learning / activity	Slides 64–74
09:50–10:00	10 min	Tea break	Break	—
10:00–11:00	60 min	Activity 5 · Fairness and explainability audit	Facilitated learning / activity	Slides 64–74
11:00–11:35	35 min	Module 6 · Data Governance and Privacy	Facilitated learning / activity	Slides 75–85
11:35–12:20	45 min	Activity 6 · Data governance controls	Facilitated learning / activity	Slides 75–85
12:20–13:20	60 min	Lunch break	Break	—
13:20–13:50	30 min	Module 7 · Generative AI Security and Copyright	Facilitated learning / activity	Slides 87–98
13:50–14:00	10 min	Tea break	Break	—
14:00–14:50	50 min	Activity 7 · GenAI threat model	Facilitated learning / activity	Slides 87–98
14:50–15:20	30 min	Module 8 · Human Oversight and Accountability	Facilitated learning / activity	Slides 100–111
15:20–15:50	30 min	Activity 8 · Oversight and governance decision	Facilitated learning / activity	Slides 100–111
15:50–16:00	10 min	Assessment briefing and PM attendance	Facilitated learning / activity	—
16:00–17:00	60 min	Written Assessment (SAQ)	Assessment	—
17:00–18:00	60 min	Case Study assessment	Assessment	—

Module and Activity Guide

Module 1 — AI Ethics Foundations • Slides 18–28

From AI capability to sociotechnical impact

Activity 1 — Map an AI Use Case and Its Stakeholders • Slide 27

Identify the system purpose, decision, stakeholders, benefits, harms and ethical tensions (A1).

Method: facilitated briefing, small-group analysis, template completion, peer challenge and evidence review.

Acceptance check: The canvas names the decision, at least six stakeholders, two benefits, four harms, three prohibited uses and an accountable owner.

Module 2 — Responsible AI Principles • Slides 29–39

Translate values into observable behaviour

Activity 2 — Turn Principles into an AI Code of Conduct • Slide 38

Translate responsible AI principles into observable duties, controls and evidence (A1, A2).

Method: facilitated briefing, small-group analysis, template completion, peer challenge and evidence review.

Acceptance check: Every principle has a mandatory duty, owner, control, frequency/trigger, evidence and failure indicator.

Module 3 — AI Governance Frameworks • Slides 41–51

Use complementary frameworks without creating checklist theatre

Activity 3 — Tier AI Risk and Build a Governance RACI • Slide 50

Apply governance frameworks to classify AI use cases and assign decision rights (A1, A3).

Method: facilitated briefing, small-group analysis, template completion, peer challenge and evidence review.

Acceptance check: All use cases have a defensible tier, minimum control profile, review cadence and unambiguous RACI.

Module 4 — AI Governance Lifecycle • Slides 52–62

Control the system from idea to retirement

Activity 4 — Complete an AI Impact Assessment and Risk Register • Slide 61

Apply the governance lifecycle to assess impact, select controls and make a recorded go/no-go decision (A1, A3).

Method: facilitated briefing, small-group analysis, template completion, peer challenge and evidence review.

Acceptance check: The assessment connects context, harms, controls, evidence, residual risk and an accountable decision with conditions.

Module 5 — Fairness and Explainability • Slides 64–74

Measure disparities and explain decisions without false certainty

Activity 5 — Audit Fairness and Explainability • Slide 73

Compute group-level fairness indicators, interpret explanations and recommend proportionate mitigations (A1, A2).

Method: facilitated briefing, small-group analysis, template completion, peer challenge and evidence review.

Acceptance check: The scorecard shows correct metrics by group, interprets harm cautiously and connects mitigation to an owner and monitoring rule.

Module 6 — Data Governance and Privacy • Slides 75–85

Make provenance, quality, access and retention governable

Activity 6 — Design Data Governance and Privacy Controls • Slide 84

Review AI data lineage, quality, privacy threats, access and retention controls (A2).

Method: facilitated briefing, small-group analysis, template completion, peer challenge and evidence review.

Acceptance check: The data plan can reconstruct provenance and sets measurable quality, access, privacy, retention and deletion requirements.

Module 7 — Generative AI Security and Copyright • Slides 87–98

Secure the data-model-application-human stack

Activity 7 — Threat-Model a Generative AI Assistant • Slide 97

Apply layered generative AI security, privacy and copyright controls (A2).

Method: facilitated briefing, small-group analysis, template completion, peer challenge and evidence review.

Acceptance check: The threat model covers data, model, application and human layers; every high-risk abuse case has multiple controls and evidence.

Module 8 — Human Oversight and Accountability • Slides 100–111

Keep meaningful human control where consequences matter

Activity 8 — Design Human Oversight and a Governance Decision • Slide 110

Select meaningful human involvement, escalation, monitoring and redress for an AI deployment (A2, A3).

Method: facilitated briefing, small-group analysis, template completion, peer challenge and evidence review.

Acceptance check: The oversight design is risk-proportionate, gives reviewers real authority, addresses automation bias and includes contestability and stop conditions.

Assessment

Written Assessment (Short-Answer Questions): 4 questions covering K1–K4, 1 hour, open book.

Case Study: 3 questions covering A1–A3, 1 hour, open book.

Learners complete assessment attendance, submit completed question papers through <https://lms-tms.tertiaryinfotech.com/>, receive feedback and sign the Assessment Summary Record.

Resources Required

- Trainer and learner slide deck
- Learner Guide
- Eight learner activity folders with prompt PDFs, mock data and checklists
- Spreadsheet and word-processing applications
- Access to the LMS/TMS portal